

Table 7: A multi-dimensional comparison of representative works in Matryoshka Representation Learning (MRL). The principle of nested, adaptable representations has been extended from embedding dimensions to model depth, token counts, and even quantization bit-widths, addressing a wide range of tasks from efficient retrieval to model interpretability.

Representative Works	Modality	Target Task	Matryoshka On	Core Innovation
MRL (Kusupati et al., 2022)	Text / Vision	Classification / Retrieval	Embedding Dimension	Foundational concept of nested embeddings.
2DMSE (Li et al., 2024c)	Text	STS / Retrieval	Embedding Dimension & Model Depth	Extends MRL to two dimensions for greater flexibility.
Starbucks (Zhuang et al., 2024)	Text	STS / Retrieval	Embedding Dimension & Model Depth	Improves 2DMSE to match individually trained models.
MatTA (Verma et al., 2025)	Generic	General Tasks	Model Depth & Width (FFN)	Teacher-TA-Student distillation for elastic student models.
M3 (Cai et al., 2024) / MQT (Hu et al., 2024)	Multimodal	VQA / Reasoning	Number of Visual Tokens	Enables adaptive visual granularity in LVLMS.
Matryoshka Re-Ranker (Liu et al., 2025g)	Text	Re-ranking	Model Depth & Sequence Length	Creates flexible re-rankers with configurable architecture.
SMEC (Zhang et al., 2025a)	Text / Vision	Retrieval / Classification	Embedding Dimension	Sequential training to mitigate gradient variance.
Matryoshka-Adaptor (Yoon et al., 2024)	Text / Multimodal	Retrieval	Embedding Dimension	Post-hoc tuning to impart Matryoshka properties.
MatMamba (Shukla et al., 2024)	Language / Vision	LM / Classification	Hidden Dimension & Heads	Generalizes MRL to State Space Model (SSM) architectures.
Matryoshka SAE (Bussmann et al., 2025)	Model Activations	Interpretability	SAE Dictionary Size	Mitigates feature splitting in Sparse Autoencoders.
Matryoshka Quantization (Nair et al., 2025)	Model Weights	Model Compression	Quantization Bit-width	Leverages nested structure of integer data types.

KaLM-Embedding-V2 (Zhao et al., 2025), EmbeddingGemma (Vera et al., 2025), jina-embeddings-v3 (Sturua et al., 2024), jina-colbert-v2 (Jha et al., 2024)) across diverse real-world tasks (Lai et al., 2024; Wang et al., 2024c; Nacar and Koubaa, 2025; Hanley and Durumeric, 2025; Fu et al., 2025). Formally, given an encoder $\mathcal{E}$, a document page d is mapped to a full embedding $\mathbf{e}_d = \mathcal{E}(d) \in \mathbb{R}^D$. MRL trains the model on a set of nested dimensions $M = \{m_1, m_2, \dots, m_k\}$ where $m_1 < m_2 < \dots < m_k = D$, such that each prefix $\mathbf{e}_d^{[m_i]} \in \mathbb{R}^{m_i}$ (the first m_i components of $\mathbf{e}_d$) is a valid representation. The training objective $\mathcal{L}_{\text{MRL}}$ sums a standard representation loss $\ell(\cdot, \cdot)$ over all nested dimensions:

$$\mathcal{L}_{\text{MRL}}(q, d^+) = \sum_{m_i \in M} w_i \cdot \ell(\mathcal{E}(q)^{[m_i]}, \mathcal{E}(d^+)^{[m_i]}) \quad (11)$$

where q is a query, d^+ is a positive document, and w_i are loss weights. This allows practitioners to truncate stored embeddings at inference time, providing a flexible dial to balance performance and efficiency. Building on this, 2DMSE (Li et al., 2024c) extends this elasticity to model depth, while frameworks like Starbucks (Zhuang et al., 2024) and MatTA (Verma et al., 2025) refine the training process to bridge the performance gap with individually trained models. The Matryoshka principle has also been generalized to new granularities and architectures: M3 (Cai et al., 2024) and MQT (Hu et al., 2024) apply it to the number of visual tokens in LVLMS; Matryoshka Re-Ranker (Liu et al., 2025g) enables flexible depth and sequence length for re-ranking; and MatMamba (Shukla et al., 2024) integrates it into State Space Models. The application of MRL has further expanded to diverse tasks such as improving feature hierarchy in Sparse Autoencoders (Bussmann et al., 2025) and enabling multi-precision models via Matryoshka Quantization (Nair et al., 2025). Finally, innovations in

the training paradigm itself, such as the sequential learning in SMEC (Zhang et al., 2025a) and the post-hoc tuning of Matryoshka-Adaptor (Yoon et al., 2024), are making the creation of these flexible embeddings more efficient and accessible. We compare the works above in Table 7.

F.4 Towards Interactive Retrieval

Integrating VDR with agentic systems holds immense potential, particularly for complex scenarios like *Deep Research*, where a query requires iterative evidence gathering from a vast candidate pool (Java et al., 2025; Chen et al., 2025e; Huang et al., 2025b; Zhang et al., 2025f). The current challenge, however, is that most agentic systems treat VDR models as passive, reactive tools rather than active, strategic partners. These agents typically operate within a predefined, reactive loop of reasoning and acting, which limits their ability to dynamically formulate or adapt a high-level retrieval strategy. For instance, while frameworks like WebThinker (Li et al., 2025f) and DeepResearcher (Zheng et al., 2025b) enable agents to interleave thought with web search actions, they often function as sophisticated *tool-executors* that reactively process information rather than as *research strategists* that proactively plan a multi-step information-seeking campaign. This limitation becomes particularly evident in complex research tasks, such as generating structured reports with dynamic outlines as demonstrated by WebWeaver (Li et al., 2025j) or creating text-chart interleaved content as in Multimodal DeepResearcher (Yang et al., 2025b), where the retrieval process itself must be strategically guided to gather diverse and structured evidence.

Future work must focus on the *co-design of agents and VDR tools to foster a more organic and strategic synergy*. A primary direction is to empower agents with explicit retrieval planning capabilities, enabling them to decompose a high-level query into a multi-step, multi-granularity re-